Derivation of the GAN objective

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a) Optimal Discriminator: $D^*(x) = \frac{p_{data}(x)}{p_{data}(x) + p_g(x)}$

Min-Max objective of GAN:

$$\min_G \max_D \mathbb{E}_{x \sim p_{data}(x)} \left[\log D(x) \right] + \mathbb{E}_{z \sim p(z)} \left[\log (1 - D(G(z))) \right]$$

p_{data} ... true data distribution

p_g ... generated data distribution

D ... Discriminator network

x ... sample from complex distribution

z ... sample from simple prior distribution

G ... Generator network

$$L(D^*, G) = \mathbb{E}_{x \sim p_{data}(x)} \left[\log D^*(x) \right] + \mathbb{E}_{z \sim p(z)} \left[\log (1 - D^*(G(z))) \right]$$

$$= \mathbb{E}_{x \sim p_{data}(x)} \left[\log \frac{p_{data}(x)}{p_{data}(x) + p_g(x)} \right] + \mathbb{E}_{z \sim p(z)} \left[\log \left(1 - \frac{p_{data}(x)}{p_{data}(x) + p_g(x)} \right) \right]$$

express Expectations

$$\overset{*}{=} \int p_{data}(x) \cdot \log \frac{p_{data}(x)}{p_{data}(x) + p_g(x)} dx + \int p_g(x) \log \frac{p_g(x)}{p_{data}(x) + p_g(x)} dx$$

$$= D_{KL}(p_{data} \| p_{data} + p_g) + D_{KL}(p_g \| p_{data} + p_g)$$

$$= 2 \cdot \left(\tfrac{1}{2} D_{KL}(p_{data} \| \tfrac{1}{2}(p_{data} + p_g)) + \tfrac{1}{2} D_{KL}(p_g \| \tfrac{1}{2}(p_{data} + p_g)) \right) - 2 \log(2)$$

$$= \underline{\underline{2 \cdot D_{JS}(p_{data} \| p_g) - 2 \log(2)}}$$

introduce factor of $\frac{1}{2}$ to apply Jensen-Shannon Divergence

$$\overset{*}{=} \int p_{data}(x) \cdot \log \frac{2\, p_{data}(x)}{2(p_{data}(x) + p_g(x))} dx + \int p_g(x) \log \frac{2\, p_g(x)}{2(p_{data}(x) + p_g(x))} dx$$

$$= \int p_{data}(x) \cdot \left(\log \frac{2\, p_{data}(x)}{p_{data}(x) + p_g(x)} - \log(2) \right) dx + \int p_g(x) \left(\log \frac{2\, p_g(x)}{p_{data}(x) + p_g(x)} - \log(2) \right) dx$$

$$= \int p_{data}(x) \cdot \log \left(\frac{p_{data}(x)}{\frac{1}{2}(p_{data}(x) + p_g(x))} \right) dx + \int p_g(x) \cdot \log \left(\frac{p_g(x)}{\frac{1}{2}(p_{data}(x) + p_g(x))} \right) dx - \log(2) \underbrace{\int p_{data}(x)\, dx}_{1} - \log(2) \underbrace{\int p_g(x)\, dx}_{1}$$

$$= D_{KL}(p_{data} \| \tfrac{1}{2}(p_{data} + p_g)) + D_{KL}(p_g \| \tfrac{1}{2}(p_{data} + p_g)) - 2 \log(2)$$